Locally projective quotients: complete written costs

Human Compression project

Private numerical candidate; distinct review pending

Two originals remain distinct: L084 is the complete two-page 2006 publisher erratum, and L095 is the full fifteen-page July 2003 arXiv v1. The accepted replacement recovers their corrected principal contracts, with all 52 source dispositions and all 145 historical rows retained. No unseen journal edition or corrected source text is substituted in the numerator.

Written papers

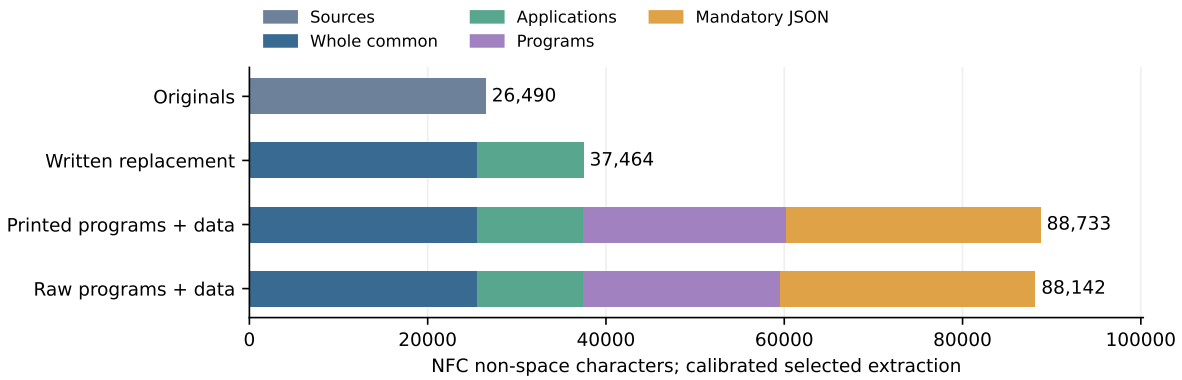
The primary unit is the number of non-whitespace characters in NFC, after exact running furniture removal and the accepted visible-symbol calibration. Titles, abstract, addresses, version banner, tables and references remain.

Complete document	Pages	Characters
L084 original	2	2,071
L095 original	15	24,419
Common proof	7	19,533
Common census tables	5	5,965
L084 application	2	4,857
L095 application	3	7,109
Combined originals	17	26,490
Whole common and applications	17	37,464

The common bundle includes the complete marked-group, geometry and census proofs. Neither application borrows an unpaid proof from its sibling or another topic. Writing $U^* = U +$ tables, the combined written comparison is

$$\frac{c(L084) + c(L095)}{c(U^*) + c(A_1) + c(A_2)} = \frac{26490}{37464} = 0.707079.$$

The written replacement is longer by 10,974 characters in this coordinate. A factor below one records that increase; it is not a compression gain.



Including the complete programs and data

All six complete mathematical programs are required finite computations. Their six-page printed appendix costs 22,695 characters; the full raw alternative costs 22,104. In *both* cases the entire required `source_tables.json` adds 28,574 characters (39,150 raw bytes). Its 145 rows are used in historical reconciliation, not subgroup enumeration. The printed code is not a substitute for these data. Thus the two alternatives are

$$\frac{c(L084) + c(L095)}{c(U^*) + c(A_1) + c(A_2) + c(\text{printed programs}) + c(\text{data})} = \frac{26490}{88733} = 0.298536,$$

$$\frac{c(L084) + c(L095)}{c(U^*) + c(A_1) + c(A_2) + c(\text{raw programs}) + c(\text{data})} = \frac{26490}{88142} = 0.300538.$$

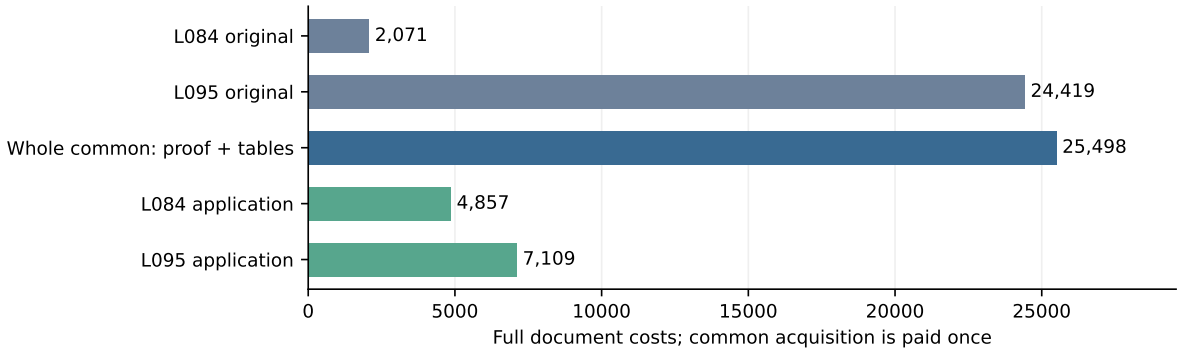
Neither row adds both code representations or prunes any file or data field.

After acquiring the whole common theory, programs and data

The conditional comparisons are source/application only after that acquisition:

$$\frac{c(L084)}{c(A_1)} = \frac{2071}{4857} = 0.426395, \qquad \frac{c(L095)}{c(A_2)} = \frac{24419}{7109} = 3.434942.$$

The prior known acquisition is 25,498 for the whole common bundle, plus the complete code and 28,574 for data: 76,767 using printed code, or 76,176 using raw code. Retained external inputs are also prerequisites; their full costs are unknown. These conditional factors do not replace the combined comparison.



Separate costs and limits

The six raw programs total 24,255 bytes. Three operational drivers total 32,220 raw bytes (29,370 primary characters). Eleven regenerated outputs total 3,470,182 bytes, including the complete computed subgroup representatives. They remain separately inventoried; outputs generated in the same computation are not charged again as independent enumeration inputs. No output-text metric or formal proof cost is inferred from storage bytes. There are no Lean files or declarations in this edition.

Classical local geometry and polytope correspondences, named finite groups, and exact GAP/ACE/NetworkX algorithms remain retained inputs. Thirteen selected runtime/model pins do not measure their full implementations or proofs. Those costs, historical software, graphics/layout and full formalization remain unknown, not zero. The unavailable historical `g.txt` is not used. No new mathematical, source, question or formal credit is claimed.

Metric and extraction sensitivity

All 30 coordinates keep the same complete boundary. “Printed+data” and “Raw+data” include the full mandatory JSON; the two conditional columns assume all prior acquisition specified above. The table rounds display only; JSON and CSV retain all 150 exact fractions.

Untrimmed extraction

Unit	Form	Written	Printed+data	Raw+data	L084 cond.	L095 cond.
Whitespace tokens	NFC	0.819493	0.527655	0.531570	0.426966	4.646223
Alphanumeric runs	NFC	0.780686	0.388078	0.390559	0.448313	4.349927
Non-space characters	NFC	0.707959	0.298966	0.300990	0.431159	3.436445
Collapsed characters	NFC	0.727686	0.327183	0.329441	0.430411	3.628196
Collapsed UTF-8 bytes	NFC	0.721622	0.331488	0.333729	0.428963	3.628085
Whitespace tokens	NFKC	0.820114	0.528056	0.531973	0.426966	4.649963
Alphanumeric runs	NFKC	0.780686	0.388078	0.390559	0.448313	4.349927
Non-space characters	NFKC	0.709770	0.299744	0.301774	0.432393	3.445444
Collapsed characters	NFKC	0.729286	0.327914	0.330177	0.431454	3.636364
Collapsed UTF-8 bytes	NFKC	0.744535	0.335913	0.338219	0.432451	3.720859

Exact furniture removed

Unit	Form	Written	Printed+data	Raw+data	L084 cond.	L095 cond.
Whitespace tokens	NFC	0.818737	0.527025	0.530685	0.422297	4.645427
Alphanumeric runs	NFC	0.779734	0.387339	0.389675	0.442748	4.348496
Non-space characters	NFC	0.707079	0.298536	0.300538	0.426395	3.434942
Collapsed characters	NFC	0.726801	0.326690	0.328906	0.425662	3.626510
Collapsed UTF-8 bytes	NFC	0.720769	0.331007	0.333207	0.424289	3.626458
Whitespace tokens	NFKC	0.819360	0.527426	0.531089	0.422297	4.649175
Alphanumeric runs	NFKC	0.779734	0.387339	0.389675	0.442748	4.348496
Non-space characters	NFKC	0.708890	0.299315	0.301322	0.427630	3.443944
Collapsed characters	NFKC	0.728402	0.327421	0.329642	0.426706	3.634684
Collapsed UTF-8 bytes	NFKC	0.743671	0.335426	0.337690	0.427731	3.719253

Furniture removed and visible symbols calibrated

Unit	Form	Written	Printed+data	Raw+data	L084 cond.	L095 cond.
Whitespace tokens	NFC	0.818737	0.527025	0.530685	0.422297	4.645427
Alphanumeric runs	NFC	0.779611	0.387278	0.389614	0.441658	4.348496
Non-space characters	NFC	0.707079	0.298536	0.300538	0.426395	3.434942
Collapsed characters	NFC	0.726801	0.326690	0.328906	0.425662	3.626510
Collapsed UTF-8 bytes	NFC	0.724142	0.332556	0.334766	0.426687	3.643151
Whitespace tokens	NFKC	0.819360	0.527426	0.531089	0.422297	4.649175
Alphanumeric runs	NFKC	0.779611	0.387278	0.389614	0.441658	4.348496
Non-space characters	NFKC	0.708890	0.299315	0.301322	0.427630	3.443944
Collapsed characters	NFKC	0.728402	0.327421	0.329642	0.426706	3.634684
Collapsed UTF-8 bytes	NFKC	0.743714	0.335446	0.337709	0.428077	3.719253

The untrimmed coordinate retains every extracted character, including running furniture. Selected variants remove only the exact erratum running header and isolated page numbers. Eight actual font classes receive 61 occurrence-bound repairs: calligraphic capitals, a correspondence envelope, proof-end squares and mapsto tails. A tail remains separate from its adjacent arrow. Printed hyphens remain literal; NFC and NFKC are distinct sensitivities.

The source’s incorrect historical counts are not changed by calibration. L095’s table rules are inline images. This is a full-document extractable-text comparison, not a complete measure of graphical construction or intrinsic proof complexity. The numerical author also authored the inputs; the accepted input review was distinct. These new calculations require their own independent review. No source extraction, GAP, census or Lean execution was repeated here.